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2016-DSE-PHY 1B-2

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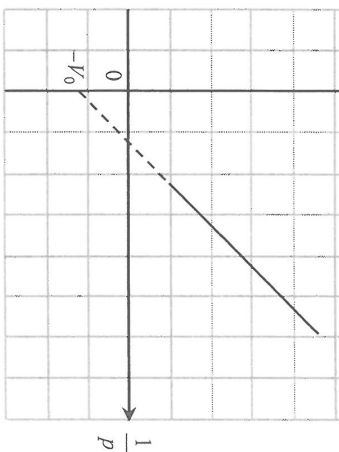
mass of gas at constant temperature. The volume V of the gas trapped is read directly from the syringe and the corresponding pressure p is measured by a data-logger via a pressure sensor.

- (a) The initial volume and pressure of the gas are $6.0 \times 10^{-5} \text{ m}^3$ and $1.0 \times 10^5 \text{ Pa}$ respectively at a room temperature of 25°C . Estimate the number of gas molecules trapped in the syringe. (3 marks)

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Figure 2.2



- (i) State ONE experimental precaution for keeping the gas temperature constant.

(1 mark)

- (ii) The straight line graph does not pass through the origin but cuts the vertical axis at $-1/V_0$ instead. Suggest what $1/V_0$ stands for. (1 mark)

- (iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure 2.2. (2 marks)

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(a) Calculate the acceleration of the lift from $t = 0$ to $t = 2$ s.

(2 marks)

The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 395 N.

(b) Match these readings with the three stages, A, B and C, of the ride (shown in Figure 3.1). Hence deduce the mass of the person. (3 marks)

A: _____

B: _____

C: _____

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(ii) Hence estimate the height of the building.

(2 marks)

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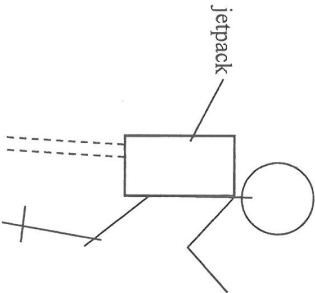
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then ejected downwards.

- (a) Referring to Figure 4.1, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton's law(s) of motion to explain why a lifting force acting on the person is produced. (3 marks)

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- (b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)



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- (ii) Water is pumped to the water jeepack at a height of 7.5 m above sea surface and then ejected from it. By considering the gain in mechanical energy of the water, estimate the minimum output power of the pump. (3 marks)

- (d) The person changes to staying 'afloat' in equilibrium at a higher position. If the speed by which water enters and is ejected from the jeepack remains the same, would the amount of water ejected per second be greater than, equal to or smaller than the result found in (c)(i) ? Explain. (Neglect the weight of the hose.) (2 marks)

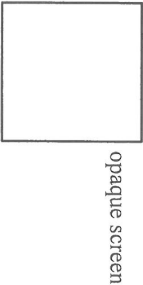
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(ii) Sketch the image on the screen seen by the observer.

(1 mark)



(b) The separation D is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance u is obtained for plotting a graph of D against u (Figure 5.2).

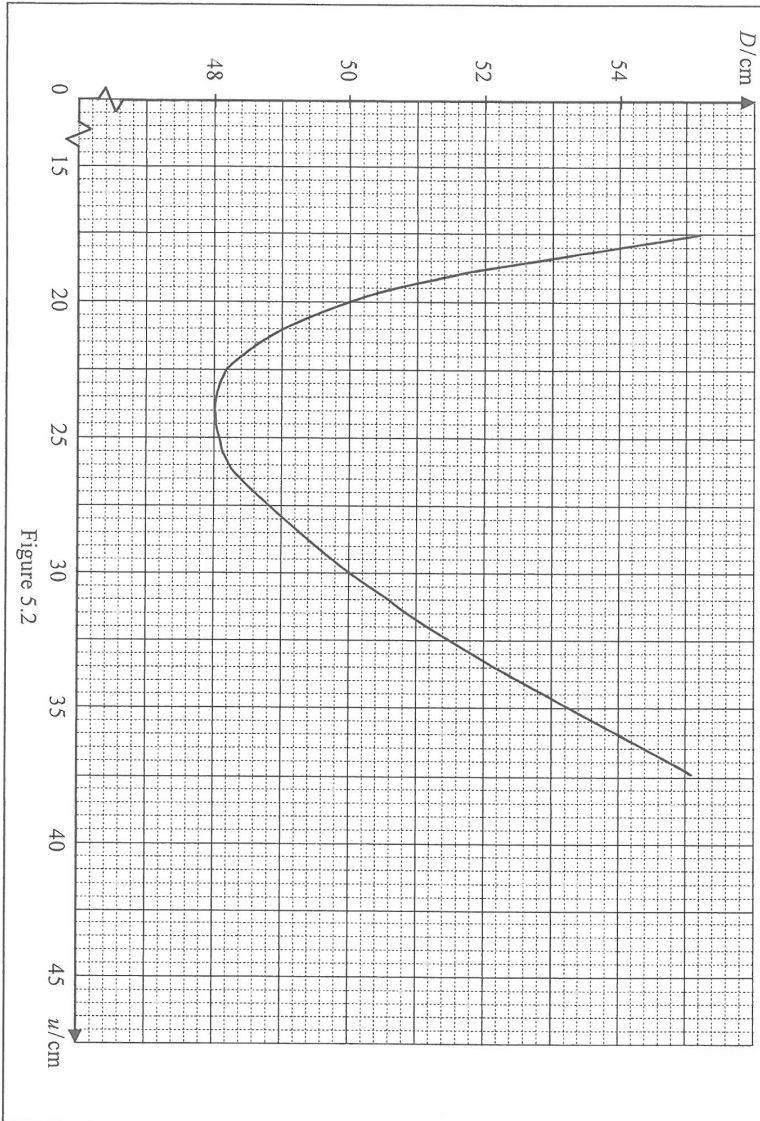
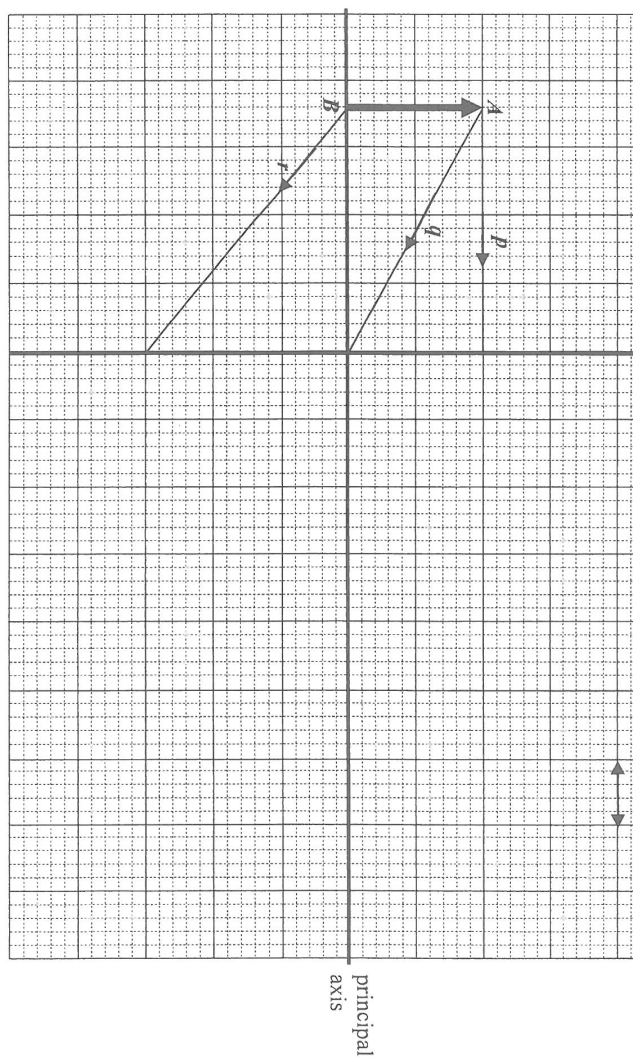


Figure 5.2

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(ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of P , q and r . (3 marks)

(iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

Focal length =

(iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

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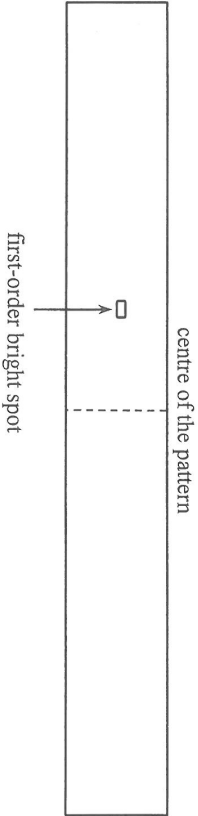
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(ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed.
(2 marks)

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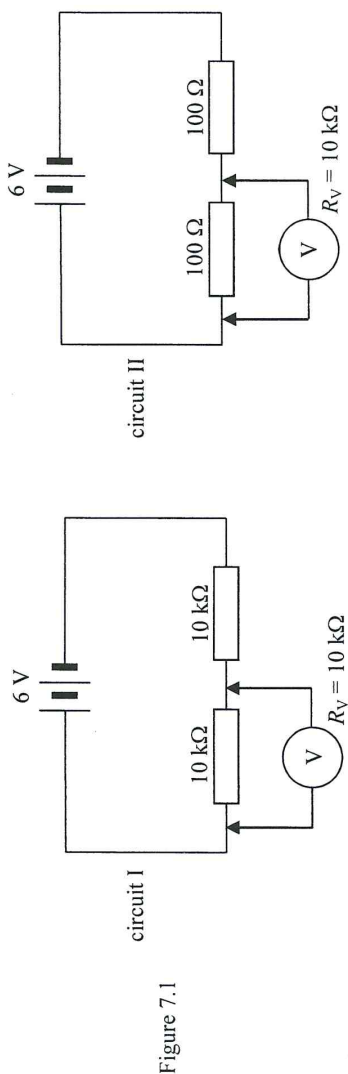
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(ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you. (2 marks)



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7. (a) The circuits in Figure 7.1 each contains two resistors connected in series with a 6 V battery of negligible internal resistance. The resistors in circuit I are $10\text{ k}\Omega$ each while those in circuit II are $100\text{ }\Omega$ each.



A voltmeter of internal resistance $R_V = 10\text{ k}\Omega$ is used to measure the potential difference across one of the resistors as shown.

- (i) What would be the respective voltmeter readings ?

(3 marks)

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- (ii) In fact, the potential difference across each resistor *before connecting the voltmeter* is 3 V in both circuits. Explain why this voltmeter gives a relatively inaccurate value for circuit I. Hence state the general principle of selecting a suitable voltmeter for such measurement.

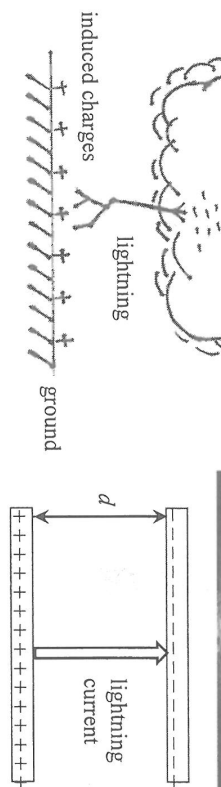
(2 marks)

(i) State which reading(s), V_m , I_m or both, do(es) **NOT** give the *true voltage* across the resistor and/or the *true current* passing through the resistor. Hence write down an equation relating R_A , R_m and R . (2 marks)

(ii) Find the percentage error associated with R_m when measuring the resistance of this resistor. (2 marks)
Given: $R_V = 10\text{ k}\Omega$, $R_A = 1\text{ }\Omega$ and $R = 10\text{ }\Omega$.

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Lightning occurs when charges accumulate in the clouds to such an extent that the electric field in the atmosphere is strong enough to cause the air to lose its insulating properties. The threshold electric field for 'breakdown' to occur is about $3 \times 10^5 \text{ V m}^{-1}$ above which electrons or ions in the atmosphere can pass through the air between clouds and the ground or between clouds and clouds. The peak current of a typical lightning bolt can reach about 30000 A. How the charges are separated and accumulated in the clouds is not fully understood yet. In most cases, negative charges are at the base of the cloud and positive charges are induced on the ground.

(a) (i) What is the meaning of 'breakdown' in the passage ?

(1 mark)

*(ii) The thundercloud's base and the ground can be modeled as two parallel plates with opposite charges. If the negative charges distributed at the cloud's base are about $d = 2 \text{ km}$ from the ground, find the potential difference between the cloud and the ground when the electric field in the atmosphere just reaches the threshold of 'breakdown'. (2 marks)

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- (ii) Explain why within the very short duration of lightning an induced current first flows in the coil in a certain direction and then reverses. Your answer should include the directions of the induced current in the coil. (3 marks)

- (iii) Among the physical quantities related to lightning, **electric field in the atmosphere, lightning current and magnetic field due to lightning**, suggest which one can be monitored so as to give fore-warning of lightning. Explain your choice. (2 marks)

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(b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to a *single decay* with half-life 4.5×10^9 years:



Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that the ratio $\frac{\text{number of Pb - 206 atoms}}{\text{number of U - 238 atoms}}$ is $\frac{2}{3}$ at present.

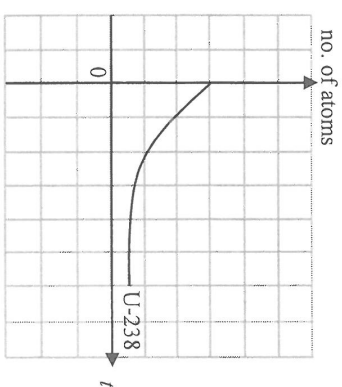
(i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

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- (iii) The graph in Figure 9.1 shows how the number of U-238 atoms in the sample varies with time t subsequently while $t = 0$ denotes the *present time*. On Figure 9.1, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

Figure 9.1



END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.